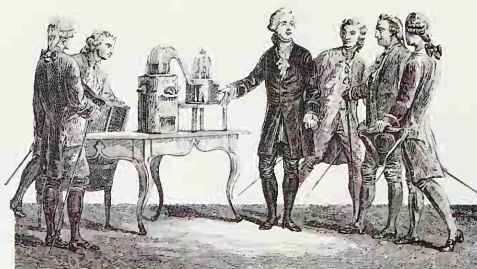


Inside the lungs

IN 1777 FRENCH CHEMIST ANTOINE LAVOISIER finally identified the nature of the fraction of air that could support life and flames. He named it oxygen. "We can state in general that respiration is but a slow combustion ... similar in all points to that taking place in a lamp or a burning candle In respiration as in combustion it is the atmospheric air which supplies the oxygen." Lavoisier used the term respiration for chemical reactions that take place within the body's cells. The word also means the physical processes of breathing. The nose, throat, windpipe, main airways and lungs, and the diaphragm are known as the respiratory system. By the time of Lavoisier, it was thought that the body used oxygen from air chemically to burn nutrients – but only in the lungs. Within 20 years Lazzaro Spallanzani showed that cellular respiration occurs in other tissues of the body. He discovered the lungs are gas exchangers: they absorb oxygen and give off carbon dioxide. This agreed with discoveries about blood circulation and the finding that blood contains both oxygen and carbon dioxide.

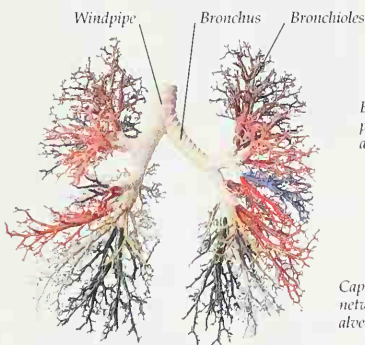


OXYGEN NAMED

Antoine Lavoisier (1743-1794) carried out many chemistry experiments that were important in the field of physiology, which he demonstrated to Parisian nobility. He named oxygen from Greek words meaning acidmaker. Throughout his life he was involved in a variety of projects, including a plan for town lighting, explosives research, the use of scientific methods in farming, and work for famine relief. He was guillotined in the aftermath of the French Revolution.

ALL-OVER RESPIRATION

Italian scientist Lazzaro Spallanzani (1729-1799) was, in turn, priest, lawyer, professor of logic and metaphysics, and professor of natural history at Pavia. He proposed that respiration did not happen just in the lungs, but in every tissue cell of the body. In his view the blood was both oxygen distributor and the collector of carbon dioxide, pumped by the heart to all body tissues.



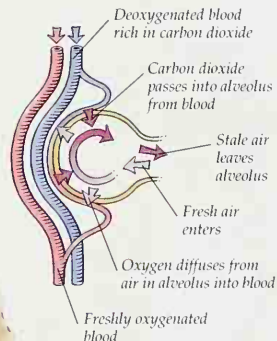
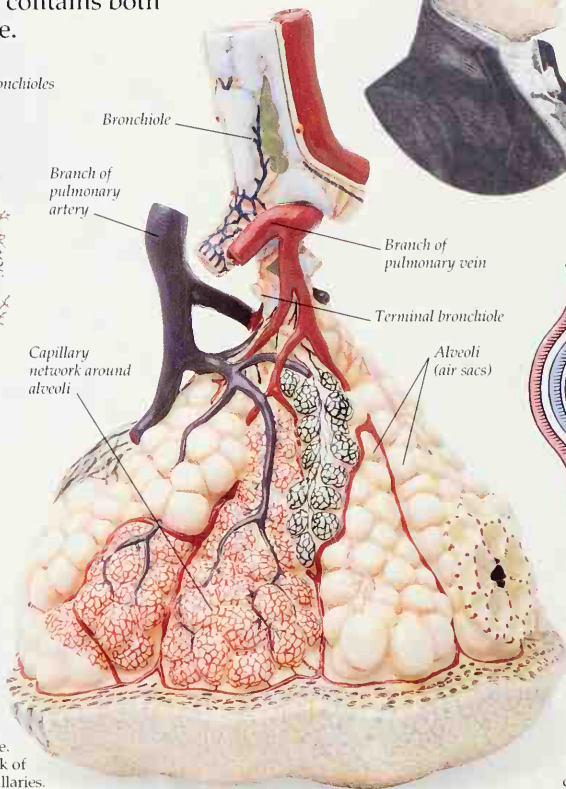
THE BRONCHIAL TREE

Inside the lungs is a branching, tree-like structure of air tubes known as the bronchial tree. The windpipe divides into two large tubes known as bronchi, one to each lung. These split a further 15 or 20 times, forming thousands of terminal bronchioles, thinner than the finest hairs.

WHERE GAS EXCHANGE HAPPENS

Each terminal bronchiole ends at a bunch of bubblelike air sacs, called alveoli. There are about 350 million alveoli in each lung. They provide a tennis-court-sized area packed into the volume of the chest to maximize gas exchange.

Around each alveolus is a network of microscopic blood vessels known as capillaries.



REFRESHED WITH OXYGEN

The lining of each alveolus is incredibly thin and coated with a film of moisture in which gases can dissolve. Oxygen from the air inside seeps through the one-cell-thick wall of the capillary into the blood, turning its color from dark to bright red. Carbon dioxide passes the opposite way.